

BILLED TO (CLIENT)

Hamza

Client
Goethe Exam Booking Bot Project

FROM (DEVELOPER)

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Full-Stack Developer & Automation Engineer
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Islamabad, Pakistan

PROJECT

Goethe-Institut Exam Booking Automation Bot

Fully automated bot for booking Goethe-Institut Pakistan German language exams (A1, A2, B1) across Karachi, Lahore, and Islamabad.

Cost Breakdown

SERVICE	DETAILS	AMOUNT (PKR)
Core Bot Engine Development Selenium automation, booking flow, form filling, CAPTCHA integration, confirmation parsing. The heart of the system that performs the actual booking.	booking_helper.py, confirmation_parser.py	80,000
Resilience & Anti-Detection Systems Selector fallback system (16 element types, 2-5 fallbacks each), circuit breaker (503/Cloudflare protection), proxy rotation with health checks, human-like delays, mouse jitter, random user agents.	selector_fallbacks.py, circuit_breaker.py, proxy_rotator.py	55,000
Backend API & Database Flask REST API with 30+ authenticated endpoints, SQLite persistence layer, student priority queue, server-side sessions, rate limiting, security headers.	webapp.py, db.py, student_queue.py	50,000
Web Dashboard (Frontend) Full admin panel with analytics cards, queue management, live logs via SSE, countdown timers, scheduling. Pure HTML/CSS/JS deployable on Netlify.	frontend/index.html	35,000

AI Assistant Integration

Flash Lite. Natural language queries for booking status, student info, logs, retry commands.	alexa.py	25,000
Monitoring & Notifications Dead man switch (heartbeat monitor), Telegram instant alerts for success/failure/crashes, email notifications, screenshot capture on booking confirmation.	deadman.py, notifications.py	20,000
Testing & Quality Assurance 66 pytest tests covering all modules: circuit breaker (12), confirmation parser (11), proxy rotator (6), student queue (8), deadman (4), selectors (3), database (8), integration (4).	Full test suite	15,000
Deployment & DevOps Docker containerization, deployment configs for Railway/Render/VPS, environment variable setup, HTTPS configuration, GUI desktop app (Tkinter).	Dockerfile, gui.py, config	10,000
Documentation & Handover Comprehensive README, architecture diagrams, deployment guides, environment variable reference, usage instructions for all interfaces (web, GUI, CLI).	README, guides	10,000

Subtotal	300,000
Tax (0%)	0
Total	PKR 300,000

Market Rate Reference

Based on 2025-2026 industry research, production-grade automation bots with similar complexity (Selenium, anti-detection, dashboard, AI integration) typically range from **\$5,000 - \$15,000 USD** (PKR 1.375M - 4.125M) for enterprise-level development. Comparable Goethe booking bots on freelance platforms are bid at PKR 42,000-125,000 for basic versions without dashboard, AI, or resilience systems. This invoice reflects **highly competitive pricing** for a production-grade, full-featured system.

~78% below global market rate

Terms & Notes

- Payment due within 30 days of invoice date
- Includes 3 months of bug fixes and minor adjustments
- Source code delivered via GitHub repository transfer
- Deployment assistance included (Railway/Render/VPS)
- MIT License - full ownership transferred to client upon payment